

Chapter two

Data and Signals in Data Transmission

Data

Definition: Information represented in a digital format (bits).

Types:

- Analog Data: Continuous signals that vary over time (e.g., voice, audio, video).
- Digital Data: Discrete signals represented by binary digits (0s and 1s) (e.g., text, images, computer data).

Cont...

Signals

Definition: The physical representation of data used for transmission.

Types:

- Analog Signals: Continuous waveforms that vary over time.
- Digital Signals: Discrete signals with two distinct levels (high and low).

Relationship Between Data and Signals

Data Conversion: To transmit data, it must be converted into a suitable signal format.

Modulation: The process of converting digital data into an analog signal for transmission over analog media.

Demodulation: The process of converting an analog signal back into digital data.

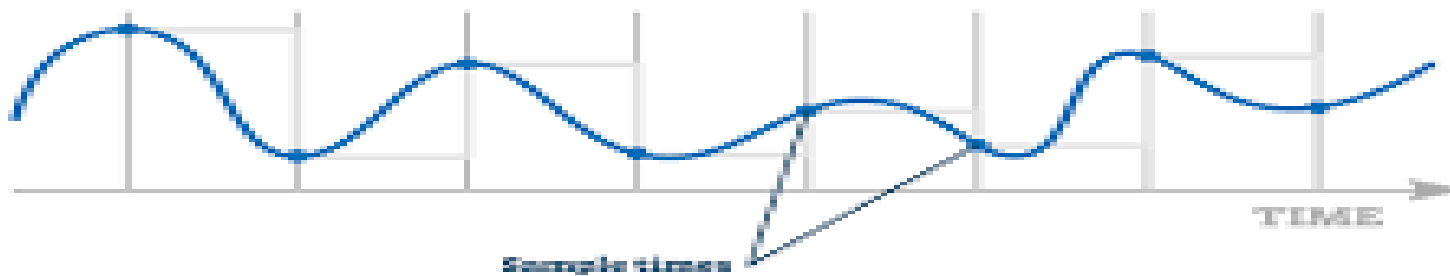
Key Concepts

- Bit Rate: The number of bits transmitted per second.
- Bandwidth: The range of frequencies available for transmission.
- Latency: The time delay between the transmission and reception of data.
- Throughput: The actual rate of data transfer, often lower than the bit rate due to overhead

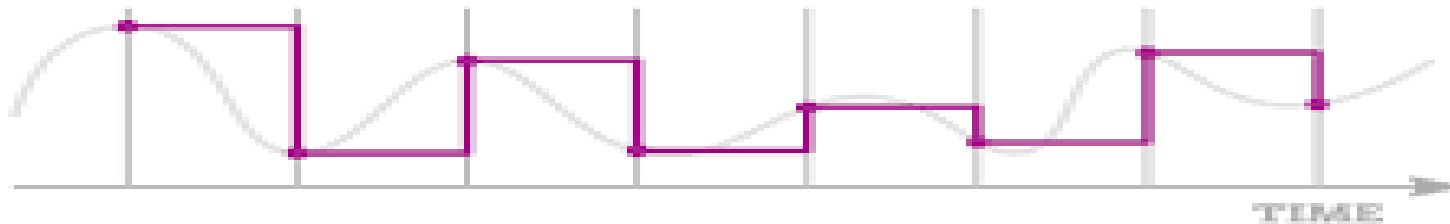
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SIGNAL

ANALOG



DIGITAL



Importance of Understanding Data and Signals

- Efficient Data Transmission: Choosing the right modulation technique and transmission medium.
- Error Detection and Correction: Implementing techniques to ensure reliable data transmission.
- Network Design: Designing networks that can handle different types of data and signals.

Physical Signaling & Encoding

Physical Signaling

Definition: The process of representing digital data as physical signals.

Key Components:

- Signal Level: The voltage or current level used to represent a binary bit (0 or 1).
- Signal Rate: The speed at which signals are transmitted, measured in bits per second (bps).
- Signal Timing: The synchronization of signals between the sender and receiver.

Encoding Techniques

Encoding techniques are used to convert digital data into a specific pattern of signals that can be transmitted over a physical medium. Common encoding techniques include:

Non-Return-to-Zero (NRZ)

NRZ-L: A binary 1 is represented by a positive voltage level, and a binary 0 is represented by a negative voltage level.

NRZ-I: A binary 1 is represented by a transition in the signal level (from high to low or low to high), and a binary 0 is represented by no transition.

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Manchester Encoding

A binary 1 is represented by a transition from low to high, and a binary 0 is represented by a transition from high to low, in the middle of each bit period

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Differential Manchester Encoding

A binary 1 is represented by a transition at the beginning of a bit period, and a binary 0 is represented by no transition at the beginning of a bit period.

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4B/5B Encoding

Groups of 4 bits of data are encoded into 5-bit codewords. This technique improves clock recovery and error detection.

Factors Affecting Physical Signaling and Encoding

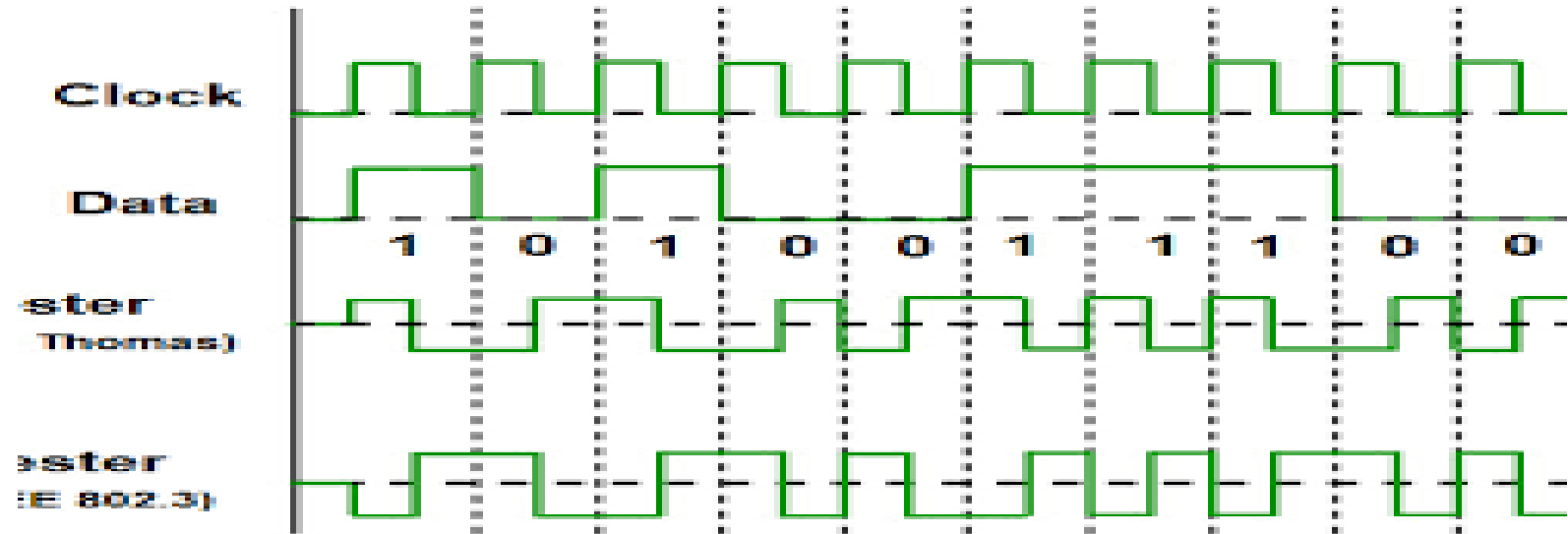
- **Transmission Medium:** The type of medium (e.g., copper cable, fiber optic cable, wireless) influences the choice of signaling and encoding techniques.

Noise and Interference: Noise can corrupt signals, so robust encoding techniques are needed to minimize its impact.

Data Rate: The desired data rate determines the signaling rate and encoding scheme.

Error Detection and Correction: Encoding techniques can be used to add redundancy for error detection and correction.

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Analog and Digital Data Transmission

Analog Data Transmission

Continuous signals that vary over time.

Examples: Voice, video, audio signals.

Challenges: Susceptible to noise and interference.

Digital Data Transmission

Discrete signals represented by binary digits (0s and 1s).

Examples: Text, images, computer data.

Advantages: More reliable, less prone to noise, and can be easily error-corrected.

Feature	Analog Data Transmission	Digital Data Transmission
Signal Type	Continuous	Discrete
Representation	Waveform	Binary digits (0s and 1s)
Noise Immunity	Less immune to noise	More immune to noise
Error Rate	Higher error rate	Lower error rate
Transmission Speed	Slower	Faster
Security	Less secure	More secure

Modulation Techniques

Process of converting digital data into analog signals for transmission over analog media.

Common Techniques:

Amplitude Shift Keying (ASK): Varies the amplitude of the carrier wave.

Frequency Shift Keying (FSK): Varies the frequency of the carrier wave.

Phase Shift Keying (PSK): Varies the phase of the carrier wave.

Quadrature Amplitude Modulation (QAM): Combines amplitude and phase shift keying.

Demodulation Techniques

Process of converting analog signals back into digital data.

Involves extracting the original digital information from the modulated signal.

Advantages of Digital Data Transmission

Error Detection and Correction: Digital data can be easily checked for errors and corrected.

Data Compression: Digital data can be compressed to reduce storage and transmission requirements.

Encryption: Digital data can be encrypted to ensure security and privacy.

Flexibility: Digital data can be easily manipulated and processed.